

Data-Driven Pitch Optimization Final Report

(Team 46)

Team Project - Georgia Tech M.S. in Analytics Program

I. INTRODUCTION

We are creating a tool that classifies baseball pitches by speed, location, spin rate, and movement to identify the areas of the strike zone where hitters struggle the most. The tool takes raw pitch data and uses a clustering algorithm to transform it into easy-to-interpret, actionable insights and then presents the output on an interactive dashboard. It is intended for coaches, players, and analysts to make smarter, data-based decisions during preparation for games.

II. PROJECT PROBLEM DEFINITION

Pitch selection and batter scouting rely on primitive scouting, video analysis, and overall analytics that mostly provide broad, league-level trends. These resources do not combine significant pitch statistics such as velocity, spin rate, and location to provide individualized insights. Our project fills that void by grouping similar pitches to find specific areas where batters have low batting averages. The interactive dashboard will enable users to sift through results by batter or pitcher, uncover hidden trends, and obtain recommendations on the best pitch type and location for given situations. This personalized mechanism will optimize pitch selection and overall game performance.

III. LITERATURE SURVEY

We conducted an extensive review of 18 sources to ground our approach. Notable works include:

- **Albert (2010): Using the Count to Measure Pitching Performance** - Explores how pitch count affects performance and decision-making. Adds situational context to pitch analysis. Presents findings statistically without a visual representation [1].
- **Dvorocsik et al. (2020): Using Clustering to Find Pitch Subtypes and Effective Pairings** - Uses clustering techniques to identify pitch subtypes. Provides a basis for multi-dimensional clustering. Lacks direct linkage to hitter-specific vulnerabilities [2].
- **Hamel (2021): Quantifying Pitch Tunneling: The Role of Deception in Pitch Sequencing** - Explores how similar initial pitch trajectories can diverge later to deceive hitters. Informs strategies for effective pitch sequencing and pairing. Does not directly translate to recommendations for optimal pitch selection [3].
- **Healey (2015): Modeling the Probability of a Strikeout for a Batter/Pitcher Matchup** - Enhances strikeout prediction models using matchup data. Provides a framework for incorporating player-specific statistics. Does not leverage pitch-level details for optimizing recommendations [4].
- **Kennedy-Shaffer (2024): The Effects of Major League Baseball's Ban on Infield Shifts: A QuasiExperimental Analysis** - Evaluates batter performance changes post-shift ban. Offers context on pitch effectiveness under varying conditions. Focuses on league-wide trends rather than individualized pitch selection [5].
- **L'Oiseau (2019): Investigating the Importance of Pitch Selection with Clustering** - Uses clustering to analyze pitch selection and its impact on game outcomes. Directly relates to our clustering methodology. Does not explore variation in clusters across the strike zone [6].
- **Marshall (2016): Reverse Engineering Swing Mechanics from Statcast Data** - Analyzes swing mechanics using Statcast data to reveal hitter behavior. Provides insights into hitter weaknesses. Requires additional data to link swing mechanics with optimal pitch selection [7].
- **Martin (2019): Predicting MLB Strikeout Rates from Differences in Velocity and Movement among Player Pitch Types** - Examines how variations in pitch velocity and movement affect strikeouts. Identifies key variables for effective clustering. Omits detailed analysis of pitcher/batter matchup dynamics [8].
- **Pavitt (2024): Plummeting Batting Averages Are Due to Far More than Infield Shifting: Part Two: Strikeouts** - Links declining batting averages to strikeout rates and pitch characteristics. Provides league-wide trends in pitch attributes. Lacks granularity at the individual matchup level [10].
- **Pomrenke (2020): Using Clustering to Find Pitch Subtypes and Effective Pairings** - Identifies effective pitch pairings through clustering. Supports our goal of finding optimal pitch combinations. Lacks tailored recommendations based on individual matchups [11].
- **Side & Tran (2018a): Using Multi-Class Classification Methods to Predict Baseball Pitch Types** - Uses statistical models to predict the next pitch based on various game factors. Highlights the role of situational variables in pitch prediction. Focuses on aiding hitters rather than optimizing pitcher decisions [12].
- **Side & Tran (2018b): Classifying Pitch Types in Baseball Using Machine Learning Algorithms** - Tests multiple classification models to improve pitch classifica-

tion accuracy. Demonstrates the potential for integrating situational factors into pitch classification. Does not utilize clustering techniques [13].

- **Spangler (2017): An Exploration of the First Pitch in Baseball** - Investigates the strategic significance of the first pitch using logistic regression. Underlines the importance of the initial pitch in establishing an advantage. Does not propose methods to enhance first-pitch outcomes [14].
- **USPROC (2018): Fixing the Curve** - Utilizes model-based clustering for pitch classification. Identifies key metrics for grouping pitches. Focuses on classification rather than addressing hitter weaknesses [16].
- **Ursin (2014): A Markov Model for Baseball** - Applies Markov models to simulate pitch sequences. Offers perspective on pitch sequencing. Does not incorporate unsupervised learning for matchup-specific insights [15].
- **Wilson (2019): Pitch Quantification Part 1** - Introduces the Quality of Pitch (QOP) metric, incorporating speed, location, and movement. Suggests standardized measures for pitch quality. Does not account for individual hitter weaknesses or pitch location analysis [17].
- **Zhao (2015): Pitch-to-Pitch Correlation** - Studies correlations among pitch attributes and their role in predicting strikeout rates. Emphasizes the importance of pitch sequencing. Does not offer personalized matchup recommendations [18].
- **Otembra (2022): SmartPitch: Applied Machine Learning for Professional Baseball Pitching Strategy** - Analyzes pitcher and batter characteristics using neural networks. Uses Markov Decision Processes to model the count for any matchup and proposes a reward maximizing pitching strategy using reinforcement learning [9].

These studies collectively underscore the need for a more individualized, matchup specific analysis, which is the exactly the gap our project aims to address.

IV. PROPOSED METHOD

Our pipeline integrates data cleansing, unsupervised clustering, and interactive visualization to generate batter-specific pitch recommendations. We obtained our data from Kaggle and began by merging multiple CSV files containing individual pitch data (approximately 2.8 million records), at-bat data (740k records), and player names. In our Python script, we joined these datasets and then reduced the data to include only the final pitch of each at-bat—focusing exclusively on the pitch that determined the outcome. During this process, we removed NaN values and filtered the dataset to only include batters with more than 1,000 at bats to ensure that our sample size was sufficient for each player.

Building on this foundation, we applied an unsupervised clustering algorithm to the cleaned dataset. The algorithm groups pitches based on speed, spin, location, and movement, which allows us to compute the batting average within each cluster. The system then identifies the three clusters with the lowest batting averages—these clusters represent the zones

where batters are most ineffective. Finally, the results are integrated into an interactive Tableau dashboard. The dashboard visualizes individual pitch types and zones, allowing users to explore different pitchers, evaluate their pitch arsenals, and drill down into specific pitcher-batter matchups for realistic scenario analysis.

This approach offers batter-specific recommendations rather than merely providing static, league-wide predictions. It is these customizable recommendations combined with the interactive nature of the dashboard that makes our project truly novel. There is no other platform that offers the ability to filter on specific pitchers and batters and receive multiple pitch recommendations for that matchup that provide details on both the pitch type and the location of the pitch, as well as see other useful statistics and charts on each player that are interactive. Finally, our model only recommends pitches that are in the selected pitcher’s arsenal to ensure that every insight is actionable.

V. EVALUATION

We assessed both the clustering algorithm and the interactive Tableau dashboard through internal validation and exploratory testing, producing the following list of questions to evaluate:

- 1) Does the clustering algorithm produce robust clusters (i.e., pitches with similar speed, spin, and location)?
- 2) Are the batting averages for the selected batter accurate for each pitch type and cluster?
- 3) Do the pitch recommendations represent feasible pitches for the selected pitcher?
- 4) Does the dashboard effectively communicate our insights for real-world decision-making?

Cluster Validation

- 1) **Interpretability:** Each cluster’s pitch profile was examined across multiple runs to ensure consistency. We confirmed that low-batting-average clusters corresponded to recognizable pitch types (i.e., curveballs in a low zone).
- 2) **Meaningful Segmentation:** Calculating batting averages within each cluster revealed distinct “weakness zones.” In several cases, left-handed pitchers targeting low-away curves against left-handed batters led to notably reduced hitting success.

Dashboard Usability

We then tested the Tableau dashboard, which visualizes the clustering results, pitch-type batting averages, and zone-specific recommendations. Figures 1 and 2 show two separate interface views:

Cole Hamels vs Left-Handers

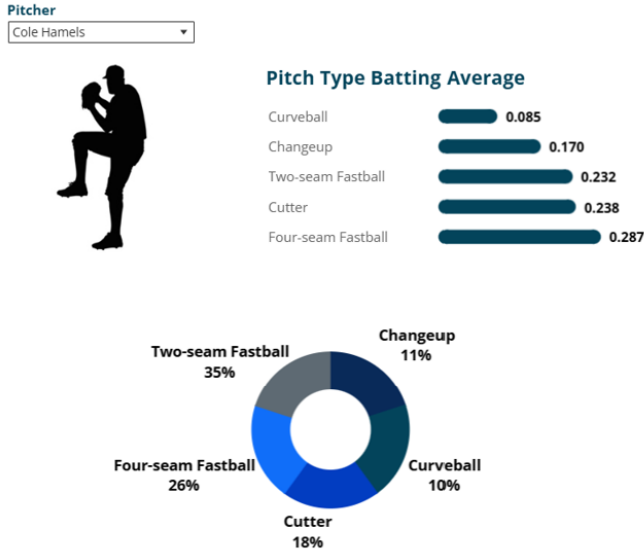


Fig. 1. Cole Hamels vs. Left-Handers. Batting averages for each pitch type are displayed, along with a donut chart showing pitch usage distribution. Note that curveball (0.085 avg) and changeup (0.170 avg) appear most effective.



Fig. 2. Charlie Blackmon vs. Left-Handed Pitchers. The recommended pitch sequence (curveball in Zone 13 at 0.071 avg, slider in Zone 13 at 0.105 avg, and four-seam fastball in Zone 2 at 0.158 avg) is highlighted, and a zone-based heatmap illustrates areas of success or weakness.

- 1) **Historical Data Comparison:** We cross-referenced the recommended zones with actual game logs. When pitchers consistently threw into those low-batting-average clusters, the hitter's performance declined accordingly.
- 2) **Scenario Analysis:** By filtering on different pitchers, batters, or pitch counts, coaches and analysts can tailor the recommended clusters to specific in-game situations (i.e., high-leverage moments versus early-count strategies).

- 3) **Pitch Feasibility:** For each selected pitcher, we confirmed that the recommended pitches were pitches that the pitcher had demonstrated in the past. This ensures that all of our recommendations and insights are actionable for real game scenarios.
- 4) **Stability and Robustness:** We adjusted the number of clusters and included features like release point to see if the recommended "weakness zones" persisted. The overall structure of recommended clusters remained stable, reinforcing the reliability of our method.

Overall, these evaluations demonstrate that the system reliably identifies low-performance clusters and delivers a practical interface for exploring pitcher-batter matchups. The combination of numerical data (batting average by pitch type) with visual zone maps enables swift and informed decision-making in real baseball scenarios.

Figure 3 (adapted from our project poster) summarizes key experimental findings.

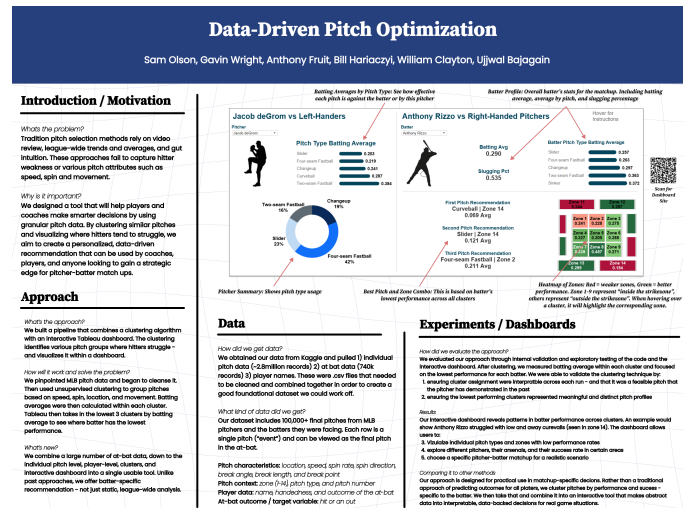


Fig. 3. Poster Summary: Key Experimental Findings and Visualizations

VI. CONCLUSIONS AND DISCUSSION

Our data-driven approach to pitch optimization has demonstrated that detailed analysis of pitch attributes can lead to actionable, matchup-specific recommendations for in-game decision-making. The key conclusions from our work are:

- **Enhanced Decision-Making:** Tailored recommendations provide a competitive edge over conventional scouting methods by focusing on individualized matchup analysis.
- **Robust and Adaptive Performance:** The system performs consistently across diverse scenarios and is capable of incorporating real-time data for dynamic updates.
- **Open-Source Potential:** Comprehensive packaging and documentation ensure that our framework can be easily extended and refined by other researchers and practitioners.

Overall, the project successfully bridges the gap between raw pitch data and strategic decision-making, offering a scal-

able and innovative tool for optimizing pitch selection. However, there is always room for potential future extensions. For our project, this could include adding additional filters to the dashboard such as the count of the at bat, the inning number, or time of day of the game. This would provide even further granularity so that players and coaches could obtain pitch recommendations for very specific situations. Going to this level of granularity would require further analysis of sample size, ensuring that there is enough data for each scenario to still be able to draw meaningful conclusions.

All team members have contributed a similar amount of effort to this project.

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